Multipole Moments

Mathew Calkins

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Abstract

Extremely rough notes on multipole moments. Maybe these will turn into something coherent, maybe not!

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1 Harmonics

Legendre polynomials arise from from large-distance moments of inverse-distance potentials. In the approximation $r \equiv |\vec{x}| \gg |\vec{x}_0| \equiv r_0$ we have

$$\begin{aligned}
\frac{1}{|\vec{x}-\vec{x}_0|} &= \frac{1}{\sqrt{r^2 + r_0^2 - 2rr_0\cos\gamma}} \\
&= \frac{1}{r}\left(1 - \frac{2r_0\cos\gamma}{r} + \frac{r_0^2}{r^2}\right)^{-\frac{1}{2}} \\
&= \frac{1}{r}\left(1 - \frac{1}{2}\left(\frac{r_0^2}{r^2} - \frac{2r_0\cos\gamma}{r}\right) + \frac{3}{8}\left(\frac{r_0^2}{r^2} - \frac{2r_0\cos\gamma}{r}\right)^2 + \cdots\right) \\
&= \frac{1}{r}\left(1 + \frac{r_0}{r}\cos\gamma + \frac{r_0^2}{r^2}\frac{3\cos^2\gamma - 1}{2} + \cdots\right) \\
&= \sum_{n=0}^{\infty} \frac{r_0^n}{r^{n+1}} P_n(\cos\gamma).
\end{aligned} \tag{1}$$

Here we intuitively think of $\vec{x}_0$ as the location of a source and $\vec{x}$ as the location of a distant probe, while γ is the (necessarily unsigned) angle between the two position vectors.

2 Spherical Harmonics

Spherical harmonics most naturally arise when solving the 3D Euclidean Laplace equation

$$\nabla\phi = 0, \quad \phi : \mathbb{R}^3 \to \mathbb{R} \tag{2}$$

by the ansatz of separation of variables in spherical coordinates

$$\phi(r,\theta,\phi) = R(r)Y(\theta,\phi) \tag{3}$$

whereupon the angular function equivalently obeys

$$-\lambda = \frac{1}{Y}\frac{1}{\sin\theta}\partial_\theta\left(\sin\theta\partial_\theta Y\right) + \frac{1}{Y}\frac{1}{\sin^2\theta}\partial_\phi^2 Y \tag{4}$$

for λ a constant also appearing in the ODE for R. Further separating

$$Y(\theta,\phi) = \Theta(\theta)\Phi(\phi) \tag{5}$$

gives the system

$$\begin{aligned} \Phi'' &= -m^2\Phi, \\ \lambda\sin^2\theta + \frac{\sin\theta}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) &= m^2 \end{aligned} \tag{6}$$

for $m \in \mathbb{R}$ an additional constant. The first equation forces $m \in \mathbb{Z}$. The second, plus demanding regularity at the poles $\theta = 0, \pi$, forces

$$\lambda = \ell(\ell+1) \text{ for } \ell \in \mathbb{Z}, \ell \geq |m|. \tag{7}$$

So we have a basis of solutions

$$Y_\ell^m(\theta,\phi) \propto e^{im\phi}P_\ell^m(\theta), \quad m = -\ell, -\ell+1, \ldots, \ell \tag{8}$$

where P_ℓ^m are the associated Legendre polynomials. In fact, this make up exactly the complete set of eigenfunctions of the spherical Laplacian

$$r^2\nabla^2 Y_\ell^m = -\ell(\ell+1)Y_\ell^m. \tag{9}$$

This notation is given in a slightly ambiguous way in some presentation, leaving it unclear where we mean the 3D Euclidean Laplacian applied to a function which just happens to be r-independent, or the Laplacian intrinsic to S^2 applied to a map $S^2 \to \mathbb{R}$? As it turns out, these agree even in higher dimensions. To see this, take the usual curvilinear expression for the Laplacian of a scalar

$$\nabla^2 f = \frac{1}{\sqrt{g}}\partial_a\left(\sqrt{g}g^{ab}\partial_b f\right) \tag{10}$$

and apply this to the Euclidean metric for $\mathbb{R}^N$ written in generalized spherical coordinates

$$ds^2 = dr^2 + r^2 d\Omega^2 \implies \sqrt{g} = r^{N-1} \tag{11}$$

Considering the cases where a, b are both angles, both radial, or a mixture, we find that

$$\nabla^2_{\mathbb{R}^N} f = \frac{1}{r^{N-1}}\frac{\partial}{\partial r}\left(r^{N-1}\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\Delta^2_{S^{N-1}} f. \tag{12}$$

So a function $f : S^2 \to \mathbb{R}$ is annihilated by the sphere Laplacian iff, when interpreted as a radius-independent function on $\mathbb{R}^N$, it is annihilated by the Euclidean Laplacian.

3 Harmonic Polynomials

Denote by $\mathrm{Hom}_\ell(\mathbb{R}^3)$ the space of all degree-ℓ homogeneous polynomial functions from $\mathbb{R}^3$ to $\mathbb{C}$, and by $\mathcal{H}_\ell(\mathbb{R}^3)$ the space of all harmonic polynomial functions from $\mathbb{R}^3$ to $\mathbb{C}$ which are homogeneous of degree ℓ.

$$\mathcal{H}_\ell(\mathbb{R}^3) = \{p \in \mathrm{Hom}_\ell(\mathbb{R}^3) \text{ with } \Delta p = 0\}. \tag{13}$$

For example, at $\ell = 0$ we get the space of all linear functions $\mathbb{R}^3 \to \mathbb{C}$.

$$\mathcal{H}_1(\mathbb{R}^3) = \mathrm{span}_{\mathbb{C}}(x_1, x_2, x_3). \tag{14}$$

At $\ell = 2$ we get the slightly more involved space

$$\begin{aligned} \mathcal{H}_2(\mathbb{R}^3) &= \{a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 a_{12}x_1x_2 + a_{13}x_1x_3 + a_{23}x_2x_3 \text{ such that } x_1 + x_2 + x_3 = 0\} \\ &= \mathrm{span}_{\mathbb{C}}\left(x_1x_2, x_1x_3, x_2x_3, x_1^2 - x_2^2, x_1^2 - x_3^2\right). \end{aligned} \tag{15}$$

How large is $\mathcal{H}_\ell(\mathbb{R}^3)$ in general? The space of degree-ℓ polynomials finds a natural basis in the space of monomials

$$\mathrm{Hom}_\ell(\mathbb{R}^3) = \mathrm{Span}_{\mathbb{C}}\left(\{x_1^{a_1}x_2^{a_2}x_3^{a_3} \text{ with } a_1+a_2+a_3=\ell\}\right). \tag{16}$$

Specifying such a monomial means specifying $a_1 \in \{0,1,\ldots,\ell\}$, then specifying $a_2 \in \{\ell-a_1, \ell-a_1+1,\ldots,\ell\}$, after which a_3 is fixed. So the number of such choices is exactly

$$\begin{aligned}\dim_{\mathbb{C}}\left(\mathrm{Hom}_\ell(\mathbb{R}^3)\right) &= \sum_{i=0}^{\ell}\sum_{j=\ell-i}^{\ell} 1 \\ &= \sum_{i=0}^{\ell}(i+1) \\ &= \frac{(\ell+2)(\ell+1)}{2}.\end{aligned} \tag{17}$$

We can compute the dimension of $\mathcal{H}_\ell(\mathbb{R}^3)$ by observing that the restriction of the Laplacian to arguments of degree ℓ

$$\Delta_\ell : \mathrm{Hom}_\ell(\mathbb{R}^3) \to \mathrm{Hom}_{\ell-2}(\mathbb{R}^3), \quad \ell \geq 2 \tag{18}$$

is surjective. As a low-order non-trivial example, x^2y^2 can be written non-uniquely as the Laplacian

$$\begin{aligned}xy &= \Delta\left(ax^4+bx^3y+cx^2y^2+dxy^3+ey^4\right) \\ &= 12ax^3+6bxy+2c(x^2+y^2)+6dxy+12y^2\end{aligned} \tag{19}$$

which has solutions $6(b+d)=1$ with all other coefficients vanishing. So rank-nullity gives the dimension as

$$\begin{aligned}\dim_{\mathbb{C}}\left(\mathcal{H}_\ell(\mathbb{R}^3)\right) &= \dim_{\mathbb{C}}\left(\ker\,\Delta_\ell\right) \\ &= \dim_{\mathbb{C}}\left(\mathrm{Hom}_\ell(\mathbb{R}^3)\right) - \dim_{\mathbb{C}}\left(\mathrm{Hom}_{\ell-2}(\mathbb{R}^3)\right) \\ &= \frac{(\ell+2)(\ell+1)}{2} - \frac{\ell(\ell-1)}{2} \\ &= 2\ell+1.\end{aligned} \tag{20}$$

3.1 Restricting to the Sphere

We introduced these homogeneous harmonic polynomials just now because there is a cute trick for turning them into a spherical harmonic. First, observe that a degree-ℓ homogeneous polynomial $p : \mathbb{R}^3 \to \mathbb{C}$ gives rise in a canonical way to $P : S^2 \to \mathbb{R}$ as

$$p(\vec{x}) = r^\ell p(\hat{x}) \equiv r^\ell p(\Omega) \equiv r^\ell P(\Omega), \quad \vec{x} = r\hat{x}. \tag{21}$$

In one of several equivalent phrasings, P is the restriction of p to S^2. Now if p is furthermore harmonic w.r.t. the Euclidean Laplacian, we can compute explicitly

$$\begin{aligned}0 &= \nabla^2_{\mathbb{R}^3}p(\vec{x}) \\ &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial(r^\ell P(\Omega))}{\partial r}\right) + \frac{1}{r^2}\Delta^2_{S^2}(r^\ell P(\Omega)) \\ &= \frac{P(\Omega)}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial(r^\ell)}{\partial r}\right) + r^{\ell-2}\Delta^2_{S^2}P(\Omega) \\ &= \ell(\ell+1)r^{\ell-2}P(\Omega) + r^{\ell-2}\Delta^2_{S^2}P(\Omega) \\ \implies \Delta^2_{S^2}P(\Omega) &= -\ell(\ell+1)P(\Omega).\end{aligned} \tag{22}$$

So P is an eigenfunction of the sphere Laplacian, hence is a linear combination of the level-ℓ spherical harmonics. So we have cooked up a linear map

$$\mathcal{H}_\ell(\mathbb{R}^3) \to \mathrm{Span}_{\mathbb{C}}\left(Y_\ell^m\right)_{|m|\leq\ell}, \quad p \mapsto p|_{S^2} \equiv P. \tag{23}$$

Some elementary algebraic geometry tells us this map is injective: if two polynomials $p, q : \mathbb{R}^3 \to \mathbb{C}$ of the same degree ℓ give rise to identical restrictions $P, Q : S^2 \to \mathbb{C}$, then their difference $p - q : \mathbb{R}^3 \to \mathbb{C}$ vanishes on the locus $x^2 + y^2 + z^2 - 1 = 0$.

$$(p - q)(\vec{x}) = 0 \text{ whenever } |\vec{x}| = 1. \tag{24}$$

But p, q have the same degree-ℓ scaling behavior by assumption, hence so does their difference, and for general $\vec{x} = r\hat{x} \in \mathbb{R}^3$ we have

$$(p - q)(\vec{x}) = r^\ell (p - q). \tag{25}$$

So $p = q$, and as claimed our restriction map is injective. On the other hand, we have already computed the dimension of the domain and codomain of this map, finding $2\ell + 1$ in each case. So our injective map must be an isomorphism, and we conclude that we have furnished a linear equivalence between the space of degree-ℓ harmonic polynomials and the degree-ℓ spherical harmonics when both are taken as complex vector spaces.

$$\therefore \mathcal{H}_\ell(\mathbb{R}^3) \simeq \mathrm{Span}_{\mathbb{C}} \left(Y_\ell^m\right)_{|m| \leq \ell} \tag{26}$$

4 Rotational Irreps

Let V_ℓ denote the $(2\ell + 1)$-dimensional representation of $SO(3)$, on which the squared total-angular-momentum Casimir acts as

$$L^2 = -\ell(\ell + 1)\mathbf{1}_{2\ell+1}, \quad \ell \in \mathbb{Z}_{\geq 0} \tag{27}$$

and the individual generators have matrix elements

$$[L_x]_{ij} = \frac{1}{2}(\delta_{i,j+1} + \delta_{i+1,j}) \tag{28}$$